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Event-driven dynamics, regional disparities, and netizen engagement: characterizing and exploring malaria discourse on Sina Weibo for enhanced health communication

Ning Wang^{1,2}, Xu Zhao^{1,2}, Chengyuan Li^{3*} and Yuanchun Mao³

Abstract

Background Malaria remains a major global health concern. Although China was certified malaria-free, imported cases have continued to emerge in recent years, and public awareness of the disease remains limited. As a major social media platform in China, Sina Weibo provides a large volume of malaria-related textual data suitable for analysing public discussion patterns and attention focus online. This study analysed discussions on Sina Weibo to help relevant institutions understand the netizens' key concerns and level of awareness regarding malaria. By characterizing discourse patterns and exploring underlying dynamics, the findings aim to inform the design of more targeted and effective health communication strategies, ultimately enhancing netizen awareness of malaria prevention and promoting a deeper understanding of relevant knowledge.

Method Using "malaria" as the keyword, relevant posts from Sina Weibo were collected between January 2019 and August 2024. Guided by the social media analysis framework, descriptive analyses and natural language processing techniques—including topic modelling, sentiment analysis, and named entity recognition—were conducted to systematically analyze netizen attention and discussion patterns related to malaria across temporal, spatial, and netizen-type dimensions.

Results A total of 81,010 malaria-related posts were collected. Netizens' attention to malaria is event-driven, with peaks around China's health assistance to Africa, public health incidents, and *World Malaria Day*, *China's National Malaria Day*, with posts related to malaria day accounting for approximately 20% of the annual total. Positive events typically trigger only short-lived peaks in attention, whereas negative events sustain engagement for a considerably longer period—approximately 1.5 times the duration of their positive counterparts. Although discussions concerning Africa have experienced a slight decline, they continue to constitute over 60% of region-specific posts annually; by contrast, discussions related to Asia and the Americas remain comparatively fragmented. Official health-related accounts focus on malaria prevention but post infrequently; personal accounts are more inclined to discuss treatment approaches and first-hand narratives of health experiences; unofficial health-related accounts show a more balanced distribution of topics, with particular emphasis on vaccines and scientific research; while organizational accounts provide institutional information on malaria control.

*Correspondence:

Chengyuan Li

teamo129@163.com

Full list of author information is available at the end of the article



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Conclusion Malaria-related discussions on Sina Weibo exhibit a clear event-driven pattern, with regional disparities possibly shaped by both disease prevalence and the visibility of international health collaborations involving China. Official health-related accounts show limited engagement. Tailored communication strategies are required to enhance netizens' awareness of malaria disease and strengthen their understanding of related knowledge.

Keywords Malaria, Social media, Netizens' attention

Background

Among the world's most prominent infectious diseases, malaria has long posed significant challenges in many developing countries. According to the latest World Malaria Report [1], there were 263 million malaria cases in 2023, up from 252 million in 2022. China has achieved remarkable progress in the elimination of malaria, earning certification as malaria-free by the WHO in 2021 [2]. Despite the successful domestic control of malaria [3], imported cases remain a significant concern [4, 5], driven by the growing number of Chinese migrant workers abroad and the rise in international business, tourism, and exchanges. Between 2011 and 2020, China's National Notifiable Diseases Reporting System (NNDRS) recorded 165 malaria deaths, of which 99.4% were imported malaria [6]. Similarly, Shanghai reported 112 imported malaria cases, primarily from Africa, with 2021 marking a decade-high, 13.39% involving severe malaria, 3.57% co-infected with COVID-19, and most patients developing symptoms within 14 days of arrival from January 2020 to December 2021 [7].

However, many studies have revealed a low level of netizen awareness and knowledge about malaria, especially in its prevention, transmission routes, and symptom recognition, leading to delays in seeking medical care, compromised treatment outcomes, and an increased risk of misdiagnosis and delayed diagnosis [8–11].

Social media, as widely accessible platforms for open interaction on a global scale, has become a cornerstone of contemporary communication [12]. Its impact extends to public health, where it plays an important role in public health campaigns, education, and advocacy, facilitating awareness, influencing policies, and advancing health initiatives [13]. In the context of malaria, an increasing number of studies are utilizing data mining technologies to analyse large-scale data from social media platforms to explore public sentiment, trending discussion topics, and updates in malaria-related knowledge [14].

Current approaches not only reveal how malaria is perceived and discussed by diverse audiences but also help identify patterns in misinformation, gaps in public understanding, and emerging areas of concern. More specifically, studies have explored the use of social media data for applications such as disease surveillance, adverse drug event (ADE) monitoring, and informing public

policy decisions. A study [15] analysed 157,951 tweets containing the *GlobalHealth* hashtag from January 1, 2014, to April 30, 2015, in conjunction with five disease-related hashtags (Malaria, HIV, TB, NCDS, NTDS), using unsupervised machine learning methods for topic categorization. The study found that prevention, control, and treatment were the primary themes of discussion, followed by advocacy and epidemiological information. It also reveals the priorities of global health advocates, funders, policymakers, and practitioners, offering valuable insights into social trends and public concerns, and highlighting the potential of social media as a public health communication platform. Another study developed an automated system that integrated Twitter data with natural language processing (NLP) techniques, including sentiment analysis and text classification, to detect ADEs related to malaria drugs, such as doxycycline. The findings suggested that Twitter, as a real-time data source, could effectively complement traditional drug monitoring systems by identifying ADEs associated with both new and existing medications, thus improving the efficiency of drug safety surveillance [16]. Furthermore, using text mining techniques, one study analysed Twitter discussions about malaria, extracting key issues and trends. These findings, in turn, provided important insights for public health decision-making [17]. In addition, a subsequent study employed sentiment analysis and topic modelling techniques to analyse tweets related to various pandemics, including malaria, cholera, Ebola, and HIV/AIDS. It revealed key emotional characteristics in public discourse—such as fear, trust, and disgust—with particularly strong expressions of trust in discussions about HIV/AIDS [18]. A more recent study used text mining and topic modelling techniques to examine Twitter posts from 2011 to 2020 regarding malaria and other parasitic diseases. The study found a consistent annual increase in the volume of tweets, alongside significant regional variations in the distribution of language and topics, offering new insights into global health issues and optimizing public health responses [19].

Despite the progress in leveraging social media data to gain valuable insights into malaria-related public health concerns and discourse, several limitations remain. Previous research has largely focused on analyzing English-language platforms, with insufficient attention paid to

users from non-English-speaking regions. In addition, these works typically offer population-level analyses, without examining detailed behaviors, preferences, or external factors that influence malaria-related discussions and information dissemination.

Unlike previous works, malaria-related posts crawled from the *Sina Weibo* platform [20] were analysed, with the platform being a prominent Chinese-language social media site that, to the best of the knowledge, has not yet been comprehensively analysed for malaria-related content. The temporal dynamics of netizens’ attention and malaria-related discussions on *Sina Weibo* over five years were examined. Meanwhile, the study also analyses thematic shifts in malaria-related discussions across different regions, revealing variations and evolving trends in netizens’ attention among these areas. Finally, the engaged netizens were categorized into four distinct groups, with each group’s main areas of focus and behavioural patterns analysed across three key aspects: malaria symptoms, medication use, and prevention.

Methods

This section provides a detailed description of the social media data and analysis methods employed in this study, guided by the structured analytical process of the Social Media Analysis (SMA) framework [21–23]. An overview of the analysis process is illustrated in Fig. 1.

Data collection

Sina Weibo was chosen as the data collection platform, given its status as one of the largest and most influential social media platforms in China [24, 25], as well as its extensive user base and accessible public data.

Using “malaria” as a search term, posts were collected from January 2019 to August 2024, capturing both historical and recent data, to ensure a comprehensive reflection of overall trends and emerging developments in malaria-related concerns and discussions.

The data includes user profiles, time data (i.e., posting time), post content, and attention-related data (e.g., number of likes, comments, and retweets). The demographic characteristics of the user base are presented in Table 1.

Furthermore, similar to prior works [26, 27], netizen categorization was performed among authenticated accounts, identifying four distinct groups: *Personal Account*, referring to accounts lacking institutional attributes and not containing keywords such as “official Weibo”, “official”, “company” in their profile or name. *Official Health-related Account*, referring to government-affiliated or authoritative health-related institutions or organizations, identified using keywords such as “CDC”, “Center for Disease Control and Prevention”, “Health Commission”, “Institute of Parasitic Disease Control”, “Hospital”, “Health Bureau”, “Health and Wellness”, “National Medical Products Administration”, and other related terms. *Unofficial Health-related Account*,

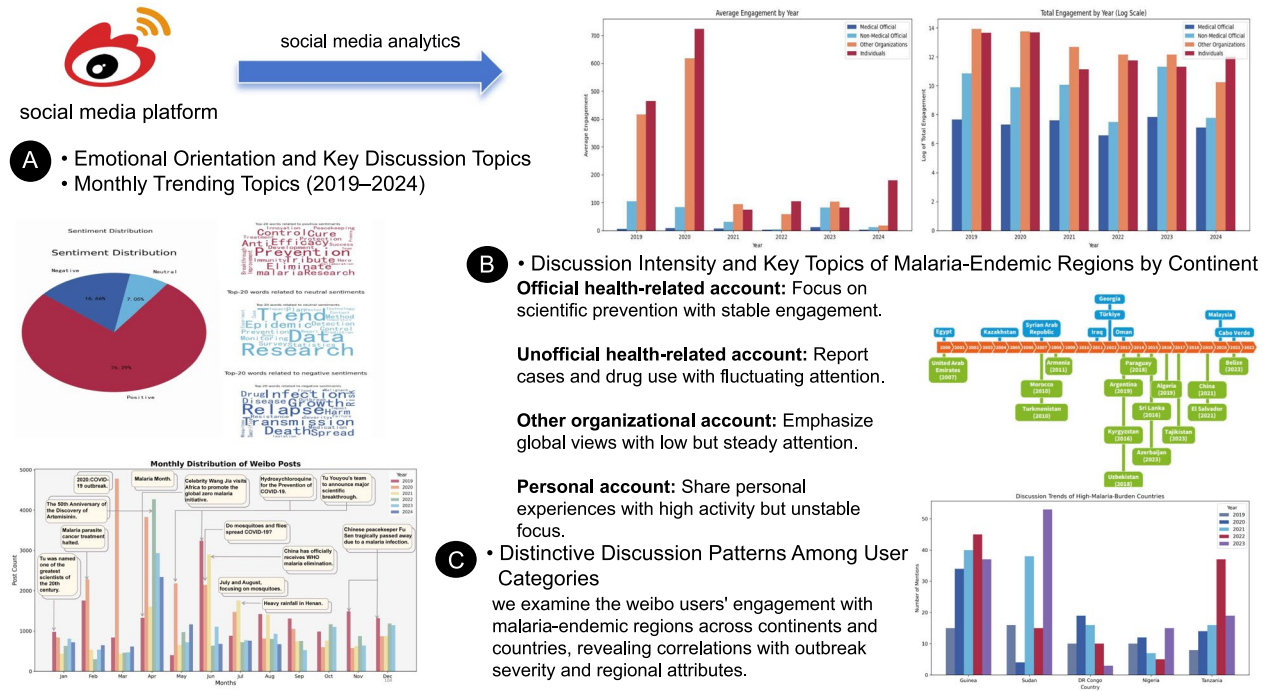


Fig. 1 An overview of the five-stage SMA framework used in the study

Table 1 Demographic characteristics of the user base

Characteristic	n	%
<i>Age group (years)</i>		
16–24	4991	13.9
25–40	13,454	37.6
50+	3255	9.1
Unknown	14,100	39.4
<i>Gender</i>		
Female	17,034	47.6
Male	13,936	38.9
Unknown	4830	13.5
<i>Geographic distribution</i>		
Jiangsu	1941	5.4
Henan	1725	4.8
Anhui	1509	4.2
Beijing	1294	3.6
Shanghai	755	2.1
Yunnan	539	1.5
Other	3019	8.4
Unknown	25,018	69.9

referring to entities involved in healthcare but not officially affiliated, including medical or biotech companies, pharmacies, platforms, and medical universities. *Other Organizational Account*, referring to companies, platforms, or institutions unrelated to the healthcare field.

Data preprocessing and cleansing

The original *Weibo* data contains both structured and unstructured data. The structured data includes meta-data such as user information, post timestamps, and engagement metrics, while the unstructured data comprises the text content of the posts. To ensure data quality, several preprocessing steps were performed: removing irrelevant or duplicate entries, and normalizing the text content by eliminating special characters, emojis. Additionally, to reduce noise and improve the efficiency of word segmentation, missing or incomplete data were filtered out. Unlike English, Chinese text does not have spaces between words, as a result, word segmentation processing was applied to the unstructured Chinese text data. The open-source Chinese word segmentation tool *Jieba* is used to tokenize *Weibo* posts and comments [28].

Topic modelling

Topic modelling can be achieved through various techniques, among which the Latent Dirichlet Allocation (LDA) model has been widely used for its effectiveness in automatically identifying semantic topics from large-scale text data [29, 30]. Therefore, the LDA model [31]

was adopted to reveal the potential semantic structure and topic patterns related to malaria in *Weibo* posts.

The *Gensim* package in Python was utilized to apply the LDA model [32]. Given the inherent challenge of LDA [31], particularly its difficulty in precisely determining the optimal number of topics and interpreting their semantic meanings [33], LDAvis [34], an interactive visualization tool, was employed to determine the optimal number of topics and to visually present and explain the identified topics.

Sentiment analysis

Social media sentiment analysis serves as a valuable tool for understanding netizens' attitudes, thoughts, and behavioural tendencies [35]. To perform the task, the *SnowNLP* library, a third-party Python tool designed for sentiment analysis on Chinese text, was used. *SnowNLP* applies natural language processing (NLP) techniques alongside a probabilistic retrieval model based on the BM25 algorithm. It classifies emotions by identifying positive and negative sentiment words, calculates the score for each emotion word within a text, and aggregates these scores to determine overall sentiment. Previous research has demonstrated that *SnowNLP* excels in sentiment analysis and corpus processing for Chinese text [21], making it an effective tool for analysis.

Named entity recognition

Named Entity Recognition (NER) [36] is a fundamental information extraction task that aims to accurately identify and classify entities such as disease symptoms, medications, and clinical terms from unstructured text. In this study, the *MING-MOE* model [37], a state-of-the-art (SOTA) deep learning framework that leverages a sparse Mixture-of-Experts (MoE) architecture with low-rank adapter modules to enhance medical multi-task learning, was adopted, demonstrating superior performance across diverse biomedical NER benchmarks.

Descriptive analysis

A descriptive analysis was conducted to summarize the general patterns of malaria-related discussions on *Sina Weibo* from January 2019 to August 2024. Posting frequencies were first analysed by year and month to identify temporal fluctuations and notable peaks in netizen engagement in malaria discussions. Subsequently, the geographical distribution of posts was examined based on malaria-endemic regions defined by the WHO. Discussion trends were compared across Africa, Asia, and the Americas to reveal cross-regional disparities in Netizens' attention and their evolution over time. Finally, netizens were categorized into four groups, including *Personal Account*, *Official Health-related Account*,

Unofficial Health-related Account, and *Other Organizational Account*, based on netizens' profile information. Posting behavioural and engagement characteristics were then analysed across the three core dimensions of malaria discourse: symptoms, medications, and prevention. This analysis provides a foundation for subsequent topic modelling and content-level interpretation.

Results

Overview of data collection

In this study, *Weibo* posts related to malaria were collected from January 2019 to August 2024. Specifically, 14,773 posts were collected in 2019, 21,437 in 2020, 12,744 in 2021, 12,776 in 2022, 11,685 in 2023, and 7595 from January 2024 to August 2024, totaling 81,010 relevant posts. Subsequently, to investigate netizen engagement on *Weibo* with malaria-endemic regions across continents and countries, the relevant posts were filtered and categorized by continent (Asia, the Americas, Africa, and Oceania), based on the *World Malaria Report 2024* published by the WHO [1]. The results reveal that 14,283 posts were related to Africa, 2190 to the Americas, 4721 to Asia, and no relevant discussions were found for Oceania. Furthermore, to analyse the focal points and behavior patterns of different netizen types on *Weibo*, verified accounts were categorized into four groups: Official Health-related Account (N = 548), Unofficial Health-related Account (N = 624), Other Organizational Account (N = 6476), and Personal Account (N = 4752).

Characterization and analysis of malaria-related Weibo discourse

Yearly analysis of malaria-related discussions

First, the key annual topics were systematically summarized and presented in Table 2.

From 2019 to 2024, malaria-related discussions on *Weibo* showed an initial upward trend, followed by a gradual decline. The number of posts peaked in 2020 at 21,437, rising from 14,773 in 2019. In the following years, the volume remained moderate but steadily decreased, with only 7595 posts recorded between January and August 2024. Public discourse on malaria also shifted thematically over time. In 2019, discussions primarily focused on artemisinin and antimalarial resistance research (5500 posts), and malaria control in Africa and China's involvement (4500 posts). In 2020, the COVID-19 pandemic triggered extensive discussions on the intersection of malaria and COVID-19 (8000 posts), followed by topics on mosquito-borne transmission and control (6500 posts). In 2021, China's certification as malaria-free became the central topic (5000 posts). In 2022, the 50th anniversary of artemisinin's discovery revitalized discussions on malaria and infectious disease control (4500 posts). In 2023, dominant themes included traditional medicine in malaria treatment (3685 posts) and historical contributions to malaria control (3500 posts). By 2024, discussions turned to malaria treatment and scientific research (2800 posts). Compared to earlier event-driven and reactive discussions, recent netizen attention has shifted toward China's achievements in malaria control, scientific advancements, and global cooperation.

Monthly peak analysis of malaria-related discussions

Subsequently, to investigate the evolving discussion trends of netizens on the topic of malaria, monthly posts from January 2019 to August 2024 were analysed, with post counts, associated key events, and sentiment patterns to identify temporal fluctuations and emerging themes. The monthly distribution is visualized in a bar chart, with major discussion events highlighted in Fig. 2,

Table 2 Topic modelling results from 2019 to August 2024

Year	Topic 1	Topic 2	Topic 3	Topic 4
2019	Artemisinin and antimalarial resistance research (5500)	Malaria control in Africa and China's involvement (4500)	Promotion of malaria vaccines (2000)	Tu Youyou's contributions and international recognition (2773)
2020	Mosquitoes, malaria transmission, and control (6500)	Malaria, COVID-19, and treatment (8000)	New roles of artemisinin and gender perspectives on poverty alleviation (3437)	China's application for WHO malaria elimination certification (3500)
2021	Global infectious disease control and vaccines (3000)	Regional health issues and international aid (2500)	China's malaria elimination certification (5000)	Treatment of infectious diseases and traditional medicine (2244)
2022	Malaria and infectious disease control (4500)	Role of traditional Chinese medicine in malaria treatment (3000)	Malaria transmission and control measures (3000)	Tu Youyou and artemisinin's research breakthrough (2276)
2023	Historical figures and control practices (3500)	Progress of global malaria vaccines (2500)	Infectious disease transmission and health risks (2000)	Traditional Chinese medicine and malaria treatment (3685)
2024	Malaria treatment and scientific research (2800)	Global perspectives and future outlook (1900)	National malaria control and public health (1400)	Mosquito transmission and disease prevention (1495)

Numbers in parentheses indicate the number of posts related to each corresponding topic

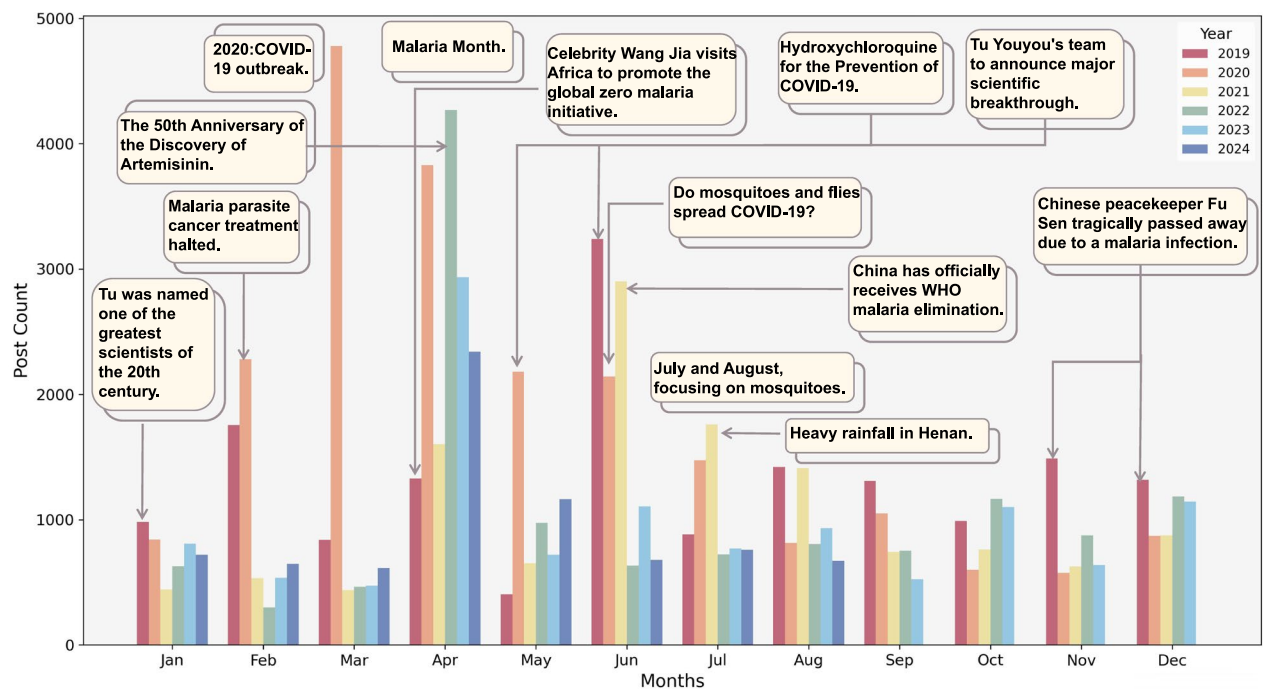


Fig. 2 Monthly distribution of Weibo posts

and the corresponding sentiment analysis results shown in Table 3.

The following key observations were received:

- Notable achievements associated with malaria or Tu Youyou consistently triggered intense Netizens' attention and positive sentiment. For example, the 50th anniversary of artemisinin discovery and Tu Youyou being named one of the greatest scientists of the 20th century elicited 98.01% and 78.20% positive sentiment, respectively.
- Following the outbreak of COVID-19, cross-topic discussions became more frequent. Topics such as “Do mosquitoes and flies spread COVID-19?” and “Hydroxychloroquine for the prevention of COVID-19” generated 91.03% and 86.84% positive sentiment, respectively, indicating high engagement and general public receptiveness. However, emotionally charged unrelated topics (e.g., “Heavy rainfall in Henan”) led to a shift, with up to 66.53% negative sentiment.
- Analysis of emotional duration shows that although scientific events trigger high initial engagement,

Table 3 Sentiment analysis results for major malaria-related events

Event description	Positive (%)	Negative (%)	Neutral (%)	Duration (days)
Tu was named one of the greatest scientists of the 20th century	78.20	16.67	5.14	8
Tu Youyou's team to announce major scientific breakthrough	88.64	4.61	6.75	7
Chinese peacekeeper Fu Sen tragically passed away due to malaria infection	17.60	80.47	1.93	13
Malaria parasite cancer treatment halted	77.12	11.24	11.64	3
Hydroxychloroquine for the prevention of COVID-19	86.84	10.22	2.94	13
Do mosquitoes and flies spread COVID-19	91.03	7.20	1.77	10
China officially receives WHO malaria elimination certification	99.13	0.63	0.24	10
Heavy rainfall in Henan	32.24	66.53	1.22	9
The 50th anniversary of the discovery of artemisinin	98.01	1.39	0.60	7
Celebrity Wang Jia visits Africa to promote global zero malaria initiative	99.01	0.59	0.40	5

Sentiment scores are expressed as percentages. *Duration* refers to the number of days the event remained a trending topic

attention tends to decline quickly. Most positive-response events, such as Tu Youyou's scientific breakthrough and the artemisinin anniversary, had a heat duration of only 7–8 days. In contrast, emotionally negative events, such as the death of a Chinese peacekeeper due to malaria (80.47% negative sentiment), sustained attention longer, with a duration of 13 days.

April malaria discussions Focused on World and National Malaria Days

As shown in Fig. 2, April consistently exhibits the highest discussion volume across all years. To further analyse these posts, topic modelling and sentiment analysis were applied, with the results presented in Table 4 and 5.

Further analysis of April posts indicates a clear concentration of activity around *World Malaria Day* (April 25th) and *National Malaria Day* (April 26th). To capture the thematic focus during this period, topic modelling was performed on posts from one week before to one week after these dates, and the results can be found in Additional File. Daily post volumes from April 23rd to 29th for the years 2019–2024 are illustrated in Fig. 3, and the corresponding topics are summarized below. Malaria-related posts around *National Malaria Day* surged from 2020, peaking at 3651 in 2022 (the 50th anniversary of artemisinin), then declined, yet remained well above 2019 levels (600), with 2649 in 2023 and 1867

in 2024. In 2019, discussions centered on prevention and drug resistance.

Moving into 2020, the emergence of *COVID-19* shifted focus to malaria–*COVID-19* cross-topic control. In 2021, conversations returned to treatment and vaccine promotion. The year 2022, marking the 50th anniversary of artemisinin, emphasized artemisinin and China's contribution. In 2023, themes included honoring Tu Youyou and control reflections. By 2024, discourse expanded to retrospective views and elimination strategies. Notably, the discussion topics during World Malaria Day closely align with the dominant themes observed throughout the year, suggesting that this period not only marks a peak in netizen engagement but also reflects the overall trajectory of malaria-related discourse.

Regional trends and thematic shifts in malaria-related discussions on Weibo

Topic modelling was conducted on *Weibo* posts that mentioned Africa, America, and Asia, and analysed the annual posting trends to uncover both region-specific concerns and common themes in malaria-related discourse. The results are presented in Fig. 4 and Tables 6.

Thematic evolution of malaria discussions in different regions

Discussions in Africa

Between 2019 and 2024, Africa consistently had the highest number of posts discussing malaria-related topics,

Table 4 Topic modelling results for April from 2019 to 2024

Year	Topic 1	Topic 2	Topic 3
2019	Transmission and health impact	Prevention, control, and treatment	Mosquito control research
2020	Intersection of malaria and the COVID-19 pandemic	Treatment and symptoms	Prevention, control, and global cooperation
2021	Prevention and elimination	Artemisinin and malaria treatment	Mosquito control and malaria transmission
2022	Contributions of artemisinin to malaria prevention and control	Research on artemisinin and its scientific value	Malaria prevention and control campaigns
2023	Achievements and challenges in prevention and control	Pandemic and control measures	Tu Youyou and the scientific breakthrough of artemisinin
2024	Prevention and control: China's role	Infectious diseases and control measures	Malaria prevention, control, and elimination

Topics are extracted from posts related to *World Malaria Day* and *National Malaria Day*, covering one week before and after April 25–26

Table 5 Sentiment analysis results for April from 2019 to 2024

Sentiment type	2019	2020	2021	2022	2023	2024
Positive (%)	71.16	67.27	65.21	83.09	75.18	77.51
Negative (%)	22.89	18.26	24.00	11.62	17.12	15.73
Neutral (%)	5.95	14.47	10.79	5.29	7.71	6.76

Sentiment scores are based on malaria-related Weibo posts from April of each year (2019–2024)

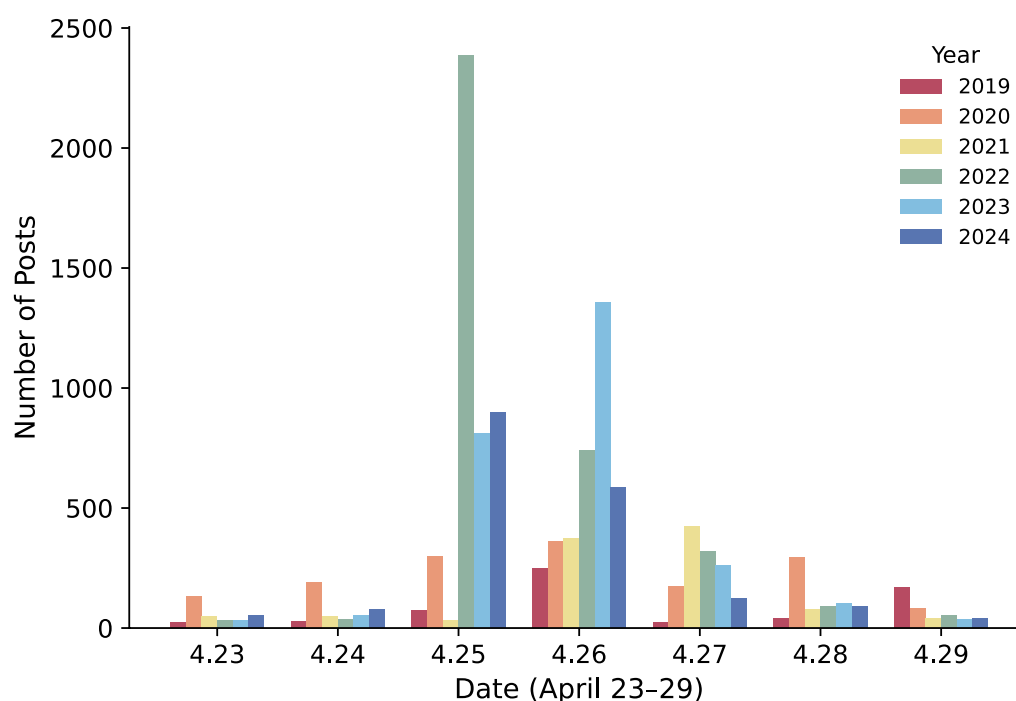
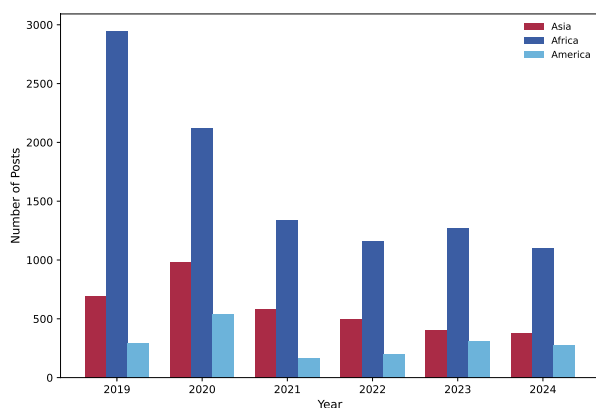
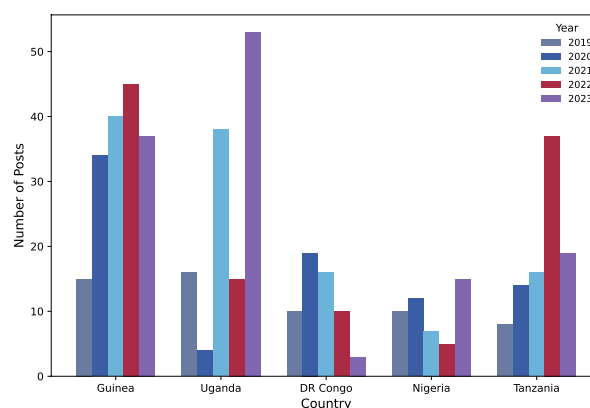


Fig. 3 Variation in daily post volume around World Malaria Day from 2019 to 2024



(a) Malaria-related discussions by region (Asia, Africa, Americas)



(b) Malaria-related discussions in high-burden African countries

Fig. 4 Trends in malaria-related discussions on *Sina Weibo*, comparing **(a)** regional engagement and **(b)** country-level focus in high-burden African countries from 2019 to 2024

although the count declined from 2945 to 1105 over the five-year period. Malaria-related discussions concerning Africa primarily focused on disease control, prevention, and regional transmission. The topics evolved over time—from peacekeeping in 2019 to imported cases and the impact of *COVID-19* in 2020, followed by discussions on transmission, prevention, and cooperation with Central Africa between 2021 and 2023. By 2024, attention shifted

to the distribution of cases across China and Africa. These thematic shifts highlight sustained concern over regional outbreaks and the importance of cross-border collaboration.

Discussions in Asia

Malaria-related discussions in Asia exhibited a fluctuating decline, rising from 696 posts in 2019 to 983 in

Table 6 Topic modelling results for malaria-endemic regions from 2019 to 2024

Year	Asia	Africa	Americas
2019	Malaria reports and the control process	Malaria control and peacekeeping	Advances and challenges in malaria treatment
2020	Interactions between malaria and the COVID-19 pandemic	Imported cases and COVID-19 control	Drug use in malaria and COVID-19 control
2021	Malaria and global epidemic connections, and control challenges	Transmission and preventive measures	Malaria–COVID-19 intersections and control measures
2022	Malaria and the control of other infectious diseases	Control and cooperation with Central Africa	Malaria and COVID-19 transmission and prevention
2023	Malaria and other infectious disease outbreak reports	Cases in China and Africa, and regional transmission	Malaria infections from overseas work
2024	Case reports and regional epidemic trends	Case distribution and transmission zones in China and Africa	Studies on malaria transmission and epidemics

Topics are extracted from malaria-related Weibo posts mentioning Asia, Africa, and the Americas from 2019 to 2024

2020, then decreasing annually to 375 by 2024. Discussion themes shifted from malaria control to malaria and *COVID-19*, and global epidemic challenges, expanding to other infectious diseases from 2022 onward, reflecting phased changes alongside the epidemic's progression.

Discussions in the Americas

Malaria-related discussions in the Americas peaked in 2020 (541 posts), declined sharply to 165 in 2021, and gradually recovered without reaching the previous peak. From 2019 to 2024, discussion themes sequentially covered advances and challenges in malaria treatment, drug use in malaria and *COVID-19* control, malaria–*COVID-19* intersection and control, transmission and prevention, overseas infections, and studies on malaria transmission and epidemics.

From 2019 to 2024, malaria discussion themes across Asia, Africa, and the Americas evolved continuously. In 2020, all three regions focused on the intersection of malaria and *COVID-19*, later expanding to transmission, outbreaks, and control. By 2024, discussions converged on case distribution and transmission patterns.

Discussion trends within Africa's high-burden malaria regions

Given the high volume of malaria-related discussions in Africa, this study conducted trend analysis and topic modelling on posts from high-burden regions based on the epidemic data from the *World Malaria Reports* (2020–2024) [38–42] (Table 7).

In Guinea, post volume steadily increased from 15 in 2019 to 45 in 2022, then slightly declined to 37 in 2023. Uganda showed greater fluctuations, dropping to 4 posts in 2020 and peaking at 53 in 2023. The Democratic Republic of the Congo exhibited a downward trend from 16 posts in 2021 to 3 in 2023. Nigeria maintained a stable volume between 10 and 15 posts, while Tanzania fluctuated, peaking at 37 posts in 2022 and declining to 19 in 2023. In 2023, Uganda recorded the highest post volume (53), followed by Guinea (37), while the Democratic Republic of the Congo and Nigeria remained consistently low. In terms of topics, Guinea emphasized international recognition (90 posts) and medical cooperation (81 posts); Uganda focused on peacekeeping-related infections (61 posts) and Chinese assistance (65 posts); the Democratic Republic of the Congo and Nigeria concentrated on malaria infections among travelers (38 and

Table 7 Discussion topics in high malaria burden countries

Country	Guinea	Uganda	Democratic Republic of the Congo	Nigeria	Tanzania
Topic 1	Medical and health cooperation and aid (81)	Malaria infection among peacekeeping personnel (61)	Malaria epidemic trends (20)	Malaria epidemic situation (14)	Malaria prevalence (20)
Topic 2	Malaria-related international honors (90)	China's contribution to malaria prevention and treatment (65)	Malaria infections among travelers (38)	Malaria infections among travelers (35)	China's medical and health assistance to Tanzania (74)

Topics are extracted from malaria-related Weibo posts in five high-burden African countries based on *World Malaria Reports* (2020–2024). Numbers in parentheses indicate the number of related posts

35 posts, respectively); and Tanzania highlighted Chinese medical aid (74 posts). Overall, Guinea and Tanzania emphasized international cooperation, whereas Uganda and the Democratic Republic of the Congo focused on local transmission and case patterns.

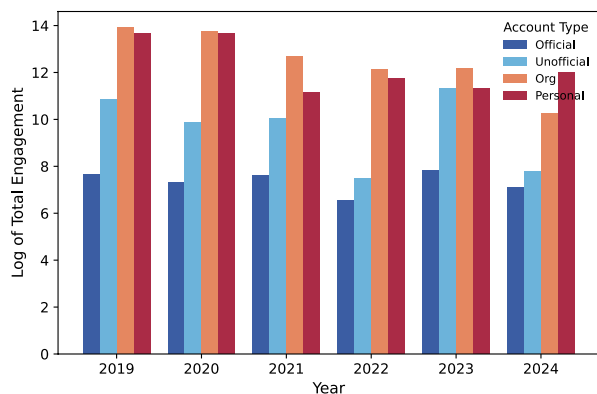
Variation in health-related discourse among different types of netizens

To further analyze discussion trends across different netizen types, changes in total attention, average attention, posting volume, and the distribution of posts across different topics [43] (Fig. 5) were examined. *Named Entity Recognition (NER)* was subsequently applied to extract entities related to symptoms, medications, and

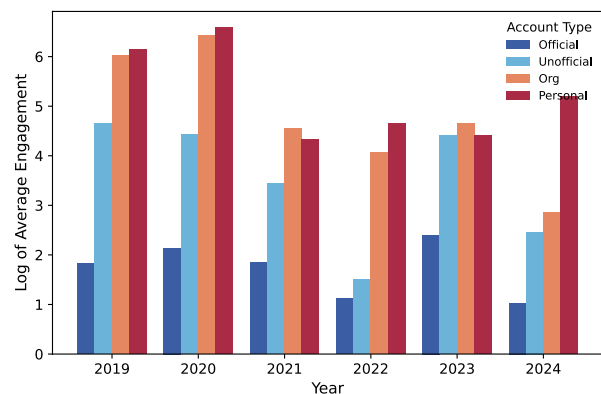
prevention in malaria-related posts, followed by topic modelling (Table 8). Detailed entity information is provided in Additional file.

Discussion pattern of different netizens

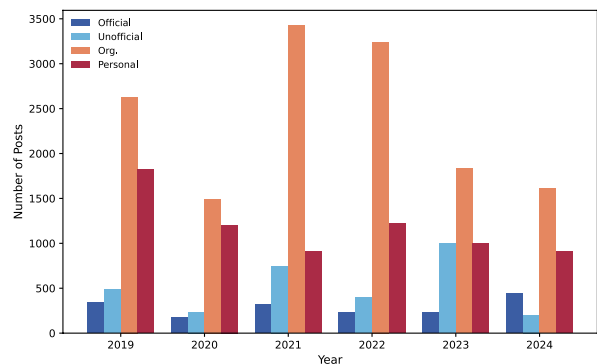
As shown in Fig. 5, *Personal Accounts* had the highest posting volume, peaking in 2020 with 1820 posts, generating approximately 868,000 engagements, followed by a continuous decline. Other *Organizational Accounts* posted the most in 2022 with 3243 posts, but maintained a low and stable average engagement. *Official Health-related Accounts* had the fewest posts in 2020 with 176 posts, while the total engagement remained relatively stable, with average engagement also staying stable across



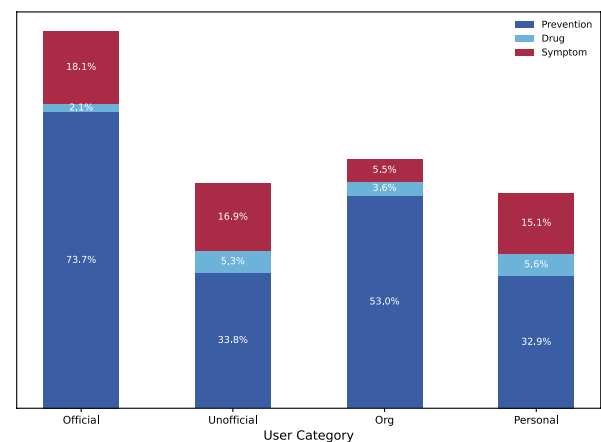
(a) Total engagement across account types (log scale)



(b) Average engagement per post (log scale)



(c) Annual post volume by account type



(d) Distribution of post categories across netizen types

Fig. 5 Comparative visualization of netizen engagement and topic focus across four types of verified accounts from 2019 to 2024: (a) Total engagement (log scale); (b) Average engagement per post (log scale); (c) Annual post volume; (d) Distribution of content categories (symptom, drug, prevention) normalized by post count

Table 8 Topic modelling results by user category

User category	Official health-related account	Unofficial health-related account	Other organizational account	Personal account
Symptoms-related topics				
Topic 1	Malaria transmission (100)	Malaria treatment (140)	Malaria transmission (199)	Malaria treatment (400)
Topic 2	Symptoms (105)	Infectious disease reporting (100)	Symptoms (300)	Symptoms (320)
Topic 3	Prevention and treatment (111)	Vaccines and symptoms (167)	Prevention efforts and epidemics (400)	Traditional Chinese medicine treatment (670)
Medication-related topics				
Topic 1	Malaria transmission (11)	Malaria treatment (35)	Malaria prevention and control (343)	Malaria treatment (133)
Topic 2	Malaria treatment (13)	Drug utilization (40)	Artemisinin (140)	Drug research (150)
Topic 3	Drug development and health (13)	Traditional Chinese medicine and scientific research (54)	Drug development and global health (100)	Vaccines and integration of traditional and modern medicine (235)
Prevention-related topics				
Topic 1	Transmission and prevention (640)	Transmission and mosquito control (400)	Prevention of mosquito-borne transmission (3000)	Relationship between malaria and environmental factors (2534)
Topic 2	Elimination and control (645)	Treatment and research (417)	Artemisinin in global malaria prevention and control (5696)	Mosquitoes and the transmission of malaria (1000)

Numbers in parentheses indicate the number of posts associated with each topic

the years. *Unofficial Health-related Accounts* showed notable fluctuations, with posting volume and engagement peaking in 2023 with 1000 posts and 53,726 total engagements, and reaching their lowest levels in 2022 with 396 posts and 6771 total engagements.

Variations in health-related discussions across netizen types

According to the topic modelling results (Table 8), the four user categories showed distinct focuses across symptoms, medications, and prevention. To further quantify these differences, statistical analysis of post distribution reveals that different netizen types exhibit varied proportions across the three themes. *Official health-related accounts* focused predominantly on prevention-related content (approximately 73.7%), while *Unofficial health-related accounts* and *Personal accounts* showed a more balanced distribution. *Other organizational accounts* also emphasized prevention (around 53.0%) but paid comparatively less attention to symptoms and medications.

For symptoms, *Personal Accounts* emphasized traditional Chinese medicine treatment (670 posts) and malaria treatment (400 posts), *Official Health-related Accounts* focused on malaria transmission (100 posts) and prevention and treatment (111 posts), *Unofficial Health-related Accounts* centered on vaccines and symptoms (167 posts), and *Other Organizational Accounts* discussed broader topics like prevention efforts and epidemics (400 posts).

For medications, *Personal Accounts* highlighted vaccines and integration of traditional and modern medicine

(235 posts) and drug research (150 posts), *Official Health-related Accounts* focused on basic treatment topics, *Unofficial Health-related Accounts* discussed traditional Chinese medicine and scientific research (54 posts), and *Other Organizational Accounts* emphasized malaria prevention and control (343 posts).

For prevention, *Other Organizational Accounts* dominated, particularly with prevention of mosquito-borne transmission (3,000 posts) and global application of artemisinin (5,696 posts); *Personal Accounts* focused on environmental factors and transmission, *Official Health-related Accounts* emphasized transmission and control, and *Unofficial Health-related Accounts* addressed mosquito control and treatment and research.

Overall, *Personal Accounts* focused on individual experiences, *Official Health-related Accounts* on scientific communication and control strategies, *Unofficial Health-related Accounts* on medications and transmission, and *Other Organizational Accounts* on macro-level prevention.

Discussion

From a temporal perspective, it was found that in 2019, discussions mainly focused on sudden events such as Tu Youyou's achievements and China's health assistance to Africa; from 2020 to 2021, the discourse gradually expanded to topics including the pandemic, vaccines, and policy developments. After 2022, attention shifted toward control mechanisms and scientific research, indicating a transition from event-driven discussions to more

stable and rational engagement with malaria-related issues. This may be due to China's shift toward malaria prevention following its elimination announcement [44], along with a decline in high-impact events after 2022 that shaped netizens' discussions. Public sentiment exhibited consistent patterns throughout this evolution. Scientific breakthroughs are typically associated with a surge in positive sentiment. Positively valenced events sustained attention only for a limited period. A small fraction of negative sentiment was attributable to concerns over the risk of malaria transmission. In contrast, negatively valenced events tended to sustain longer engagement. Research has shown that negative emotions are more likely to be retained in long-term memory compared to positive emotions, which may help explain their capacity to sustain prolonged netizens' attention [45]. Notably, some discussions not directly related to malaria also attracted considerable attention, often centered on perceived transmission risks or associations with other public health incidents. Each April, discussions around *World Malaria Day* and *China's National Malaria Day* show clear periodic peaks, focusing on representative figures, commemorative milestones, and national achievements. These commemorative days function as symbolic triggers, with their thematic content closely aligned with dominant topics throughout the year, underscoring their agenda-setting and mobilizing roles. This pattern echoes the observation by Anderson et al. [46] that *World Malaria Day* serves as a critical moment for fostering netizens' engagement. Based on the three analytical dimensions above, the following communication strategies and policy recommendations were proposed: First, posts mentioning public figures such as Wang Jia and Tu Youyou tend to attract significantly more netizens' attention. Therefore, leveraging celebrity influence may enhance the visibility and dissemination of malaria-related knowledge—for example, through short videos, public service campaigns, or collaborations with health education platforms to communicate core information on malaria, thereby deepening netizens' understanding and strengthening awareness of malaria disease. In addition, netizens often express strong national pride in response to scientific achievements and China's health assistance in Africa. These emotionally resonant events can serve as effective entry points for improving awareness of malaria disease and stimulating online engagement. By integrating real-time netizen discourse, scientific advancements, and international health assistance, they can facilitate thematic communication efforts that deepen netizens' understanding of and attention to malaria. Second, negatively valenced events typically sustain netizens' attention over longer periods. This underscores the need for timely and proactive responses from

official sources to clarify misinformation and mitigate the spread of rumours, thereby preventing unnecessary anxiety among netizens. Although scientific breakthroughs generate overwhelmingly positive sentiment—exceeding 90%—threat-oriented topics such as imported malaria receive considerably less attention. This suggests a potentially over-optimistic perception among netizens. Although China has officially declared the elimination of indigenous malaria in recent years, imported cases continue to occur and pose a potential risk of transmission. Future communication efforts could focus on strengthening risk awareness, emphasizing the dangers of imported malaria, and improving the vigilance of netizens. Third, discussions around *World Malaria Day* and *China's National Malaria Day* consistently peak each year, representing key opportunities for targeted health communication. As netizens frequently share content related to Tu Youyou and other scientific accomplishments during these periods, public health institutions could align outreach efforts with these narratives by producing educational videos, themed reports, or interactive campaigns to strengthen netizens' understanding of malaria disease and foster sustained scientific engagement.

Africa-related topics have maintained a prominent presence on *Weibo*, likely driven by the sustained momentum of China's health assistance to Africa. In contrast, discussions related to Asia appear more scattered, possibly due to the absence of high-profile aid initiatives or cross-border cooperation events, as well as the relatively low number of malaria cases in the region. Since 2020, discourse on the Americas has been largely embedded within the COVID-19 context, with malaria mentioned primarily as a secondary issue rather than a standalone topic—potentially due to its perceived similarity to COVID-19 in netizens' discourse. Despite being the country with the highest number of malaria cases globally since 2019, *Nigeria* has received minimal attention on *Weibo*, particularly between 2021 and 2022. This limited visibility may be attributed to its exclusion from China's key malaria aid initiatives and the absence of cooperation events capable of stimulating media and netizens' engagement [47]. In contrast, *Uganda* experienced a notable surge in attention in 2023, having attracted little discussion in 2020. This peak coincided with a worsening malaria situation locally [48], and may also have been influenced by China's emergency donation of 500,000 doses of antimalarial drugs in early 2024 [49]. The media coverage surrounding this aid likely contributed to heightened netizens' discourse at the end of 2023.

Africa-related topics have consistently attracted high attention on *Weibo*, largely due to sustained netizens' interest in China's health assistance efforts. Public health institutions may further leverage this engagement to

enhance malaria awareness. In contrast, discussions concerning Asia remain fragmented, likely reflecting the absence of high-profile aid programs or major cross-border initiatives. Strategic emphasis on existing regional collaborations and disease control achievements could foster a more coherent understanding of malaria risks in Asia among netizens. In addition, several Asian countries, such as India [50], continue to report malaria cases. Frequent travel and labor migration between these countries and China increase the risk of imported malaria, highlighting the need to enhance netizen awareness of associated risks and reinforce preventive vigilance. Since 2020, discourse related to the Americas has often been embedded within broader *COVID-19* narratives, with malaria marginalized or conflated due to perceived similarities. Future health communication could clearly differentiate malaria from other diseases, highlight their distinct prevention strategies and risk profiles, and raise awareness of imported malaria cases from the Americas to maintain netizen vigilance.

Furthermore, distinct functional roles were evident across netizen types on the platform. *Official Health-related Accounts* primarily focused on public health education and disease prevention. However, between 2019 and 2024, only 1725 posts were published, reflecting limited activity and low levels of user engagement, which constrained the overall effectiveness of their health communication. In contrast, *Unofficial Health-related Accounts* and *Personal Accounts* featured more diverse content, including traditional medicine and individual experiences. *Personal Accounts* published a total of 7068 posts and received as many as 1,860,121 likes, demonstrating the highest level of engagement and reflecting the broad participation of the netizens in health communication. *Unofficial Health-related Accounts* published 3,066 posts during the same period, also exhibiting a notable degree of communicative vitality. *Other Organizational Accounts* predominantly addressed control measures and international health assistance. A total of 14,227 posts were published, garnering over 2,991,196 likes, demonstrating their strong communicative influence. Although *Official Health-related Accounts* possess high institutional authority, their limited interactivity diminishes their influence in health communication.

Despite their institutional authority, *Official Health-related Accounts* currently demonstrate low posting frequency and limited engagement, which undermines their communication impact. These accounts could enhance both content quantity and quality by adopting interactive formats such as infographics, educational videos, and live Q&A sessions to improve netizens' engagement. *Unofficial Health-related Accounts* and *Personal Accounts* frequently share content related to traditional medicine

and personal experiences, though the reliability of such information is often uncertain. *Official Health-related Accounts* are advised to proactively engage with these discussions by offering evidence-based clarifications, expert interpretations, and clear guidance to ensure the dissemination of reliable health information. For *Other Organizational Accounts*, diversifying content—such as featuring local success stories, behind-the-scenes views of aid efforts, and engaging multimedia updates—may help humanize institutional narratives, increase netizens' interest, and foster more attention to malaria prevention and global health cooperation.

However, certain limitations and barriers to the direct adoption of the approach for addressing specific challenges in the communication strategies of official accounts were acknowledged. Though it offers a valuable tool to detect negative sentiment trends, highlight instances where misinformation may amplify public concern, and identifying regional or national outreach blind spots to prioritize attention, it does not directly correct misinformation or prevent negative sentiment, and it cannot by itself classify specific exposure types such as tourists or migrant workers. These aspects require integration with domain expertise and complementary data sources. Nevertheless, the approach provides a solid foundation that official departments can easily adopt or commission. Its clearly defined workflow and modular core components reduce reliance on extensive domain expertise, making it easier to adapt across diverse operational contexts. Furthermore, the cost-effectiveness and real-time analytical capability enhance its practicality, enabling official accounts to strengthen social communication, mitigate misinformation risks, and ultimately improve public health awareness with ease.

Conclusion

This study systematically examined the dissemination patterns of malaria-related discourse on Chinese social media using data from the *Sina Weibo* platform. The study finds that malaria-related discussions on *Sina Weibo* follow a highly event-driven pattern, with netizens' attention peaking in 2020 and steadily declining in the following years. Regional disparities in malaria-related discussions may be influenced by both the prevalence of the disease and the visibility of international health collaborations. Different types of netizens play distinct roles in health communication. Although unofficial accounts provide diverse content, the reliability of their information is often questionable. Therefore, official accounts need to ensure content reliability and become more active to effectively guide the dissemination of health information. The findings suggest that public health authorities are encouraged to enhance the

proactiveness and responsiveness of official accounts, tailoring content and engagement strategies to different user groups in order to improve netizen awareness of malaria risks and prevention. Given the demographic characteristics of *Weibo* users, this study may not fully capture high-risk populations such as overseas migrant workers. Future efforts should therefore explore social media platforms such as WeChat and adopt more tailored communication strategies for underserved groups.

Supplementary Information

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Supplementary material 1

Author Contributions

W.N. was responsible for the conceptualization and design of the study, developed the methodology, carried out pattern recognition and extraction tasks, and drafted the original manuscript. Z.X. contributed to the methodology and pattern extraction, participated in the drafting of the original manuscript, and reviewed the final version. M.Y.C. conducted the literature review and provided technical references, and also contributed to the revision of the manuscript. L.C.Y. contributed to the methodology and was responsible for reviewing and editing the manuscript. All authors contributed to the interpretation of the results, reviewed, and approved the final manuscript prior to submission.

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Availability of data and materials

The datasets used in this study are available from the corresponding author upon reasonable request.

Code availability

The code for this project is fully available and can be accessed at the following GitHub repository: https://github.com/Nemophilist-art/malaria_data_mining.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no Conflict of interest.

Author details

¹School of Artificial Intelligence and Computer Science, Jiangnan University, 1800 Lihu Avenue, Wuxi 214121, Jiangsu, China. ²Engineering Research Center of Intelligent Technology for Healthcare, Ministry of Education, Wuxi 214121, Jiangsu, China. ³National Health Commission Key Laboratory of Parasitic Disease Control and Prevention, Jiangsu Provincial Key Laboratory on Parasite and Vector Control Technology, Jiangsu Institute of Parasitic Diseases, Wuxi 214064, Jiangsu, China.

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